Supplementary Material: Proofs, Algorithms and Additional Experiments for "Steering Through the Trees: Physical Lie Vector Field Evasion of Tree Ensembles in Network Intrusion Detection"

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S1. Proofs

Proposition 1 and Theorem 1 are stated in Section 3.2 of the main paper; the notation is that of Section 3.1.

Proof of Proposition 1. We evaluate the vector bracket $[X, Y] = J_Y X - J_X Y$ coordinate-wise across all six distinct generator pairs:

1. **Non-Derived Coordinates $\{0, 1, 2, 3, 4, 5, 6, 7, 11\}$:** For these coordinates, the components of all four generators are either constants $(0, 1, L_0, H_0)$ or depend solely on their own coordinate (e.g., $\vec{v}_{\text{win}}$ depends only on x_7). Since no generator modifies an independent variable on which another generator depends, the Jacobian products vanish identically: $(J_Y X)_k = 0$ and $(J_X Y)_k = 0$ for all $k \notin \{8, 9, 10\}$.

2. **Temporal Dilation and Payload Padding ($X = \vec{v}_{\mathbf{time}}, Y = \vec{v}_{\mathbf{pad}}$):** X has non-zero entries in rows $6, 8, 9$ ($X_6 = 1, X_8 = -x_8/x_6, X_9 = -x_9/x_6$); Y has non-zero entries in rows $4, 8, 10$ ($Y_4 = 1, Y_8 = 10^6/x_6, Y_{10} = 1/x_2$). In row 8:

$$(J_Y X)_8 = \frac{\partial Y_8}{\partial x_6} X_6 = -\frac{10^6}{x_6^2} \cdot 1 = -\frac{10^6}{x_6^2} \tag{1}$$

$$(J_X Y)_8 = \frac{\partial X_8}{\partial x_4} Y_4 + \frac{\partial X_8}{\partial x_8} Y_8 = 0 + \left(-\frac{1}{x_6}\right)\left(\frac{10^6}{x_6}\right) = -\frac{10^6}{x_6^2} \tag{2}$$

   Subtracting yields $[X, Y]_8 = -\frac{10^6}{x_6^2} - \left(-\frac{10^6}{x_6^2}\right) \equiv 0$. In rows 9 and 10, all corresponding partial derivatives vanish, so $[X, Y] = \mathbf{0}$.

3. **Temporal Dilation and Chaff Injection ($X = \vec{v}_{\mathbf{time}}, Z = \vec{v}_{\mathbf{chaff}}$):** Z has entries $Z_2 = 1, Z_4 = L_0, Z_8 = L_0 \times 10^6/x_6, Z_9 = 10^6/x_6, Z_{10} = (L_0 - x_{10})/x_2, Z_{11} = H_0$. In row 8: $(J_Z X)_8 = -L_0 \times 10^6/x_6^2 = (J_X Z)_8$. In row 9: $(J_Z X)_9 = -10^6/x_6^2 = (J_X Z)_9$. In row 10: $(J_Z X)_{10} = 0 = (J_X Z)_{10}$. Thus $[X, Z] = \mathbf{0}$.

4. **Payload Padding and Chaff Injection ($Y = \vec{v}_{\mathbf{pad}}, Z = \vec{v}_{\mathbf{chaff}}$):** In ambient $\mathbb{R}^{12}$ coordinates, $Y_{10}(x) = 1/x_2$ and $Z_{10}(x) = (L_0 - x_{10})/x_2$. Evaluating the ambient cross-Jacobian products in row 10:

$$(J_Z Y)_{10} = \sum_{k=0}^{11} \frac{\partial Z_{10}}{\partial x_k} Y_k = \frac{\partial Z_{10}}{\partial x_{10}} Y_{10} = \left(-\frac{1}{x_2}\right)\left(\frac{1}{x_2}\right) = -\frac{1}{x_2^2} \tag{3}$$

$$(J_Y Z)_{10} = \sum_{k=0}^{11} \frac{\partial Y_{10}}{\partial x_k} Z_k = \frac{\partial Y_{10}}{\partial x_2} Z_2 = \left(-\frac{1}{x_2^2}\right) \cdot 1 = -\frac{1}{x_2^2} \tag{4}$$

Subtracting gives $[Y, Z]_{10} = (J_Z Y)_{10} - (J_Y Z)_{10} = -\frac{1}{x_2^2} - \left(-\frac{1}{x_2^2}\right) \equiv 0$. In rows 8 and 9, all entries vanish identically because $Y_6 = Z_6 = 0$ while the rate entries Y_8, Z_8, Z_9 depend spatially only on duration x_6, ensuring $(J_Z Y)_k = (J_Y Z)_k = 0$ for $k \in \{8, 9\}$.

5. **Pairs Involving Window Morphing $\vec{v}_{\mathbf{win}}$:** Because $\vec{v}_{\text{win}}$ acts exclusively on x_7, which is neither read nor modified by $\vec{v}_{\text{time}}, \vec{v}_{\text{pad}}, \vec{v}_{\text{chaff}}$, all cross-Jacobians vanish identically: $[\vec{v}_k, \vec{v}_{\text{win}}] = \mathbf{0}$ for all $k \in \{1, 2, 3\}$.

Hence, $[\vec{v}_i, \vec{v}_j] = \mathbf{0}$ identically for all pairs $i, j \in \{1, 2, 3, 4\}$. □

Corollary S1 (Frobenius Integrability and Commuting Flows)**.** *On the regular open subdomain where the four generators are linearly independent (specifically where $x_6 > 0$, $x_2 \geq 1$, and $x_7 \neq W_{target}$ for TCP), the distribution $\Delta = \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ has constant rank 4. Because $[\vec{v}_i, \vec{v}_j] \equiv \mathbf{0}$, Δ is involutive and completely integrable by Frobenius' Theorem. At points where $x_7 = W_{target}$, $\vec{v}_4 = \mathbf{0}$ and Δ smoothly reduces to the 3-dimensional involutive distribution spanned by $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. Furthermore, the generated continuous flow maps commute:*

$$\varphi_i^{s_1} \circ \varphi_j^{s_2} = \varphi_j^{s_2} \circ \varphi_i^{s_1}, \quad \forall s_1, s_2 \geq 0 \tag{5}$$

forming an Abelian local Lie semigroup action (commuting local flows $\varphi_i^{s_1} \circ \varphi_j^{s_2} = \varphi_j^{s_2} \circ \varphi_i^{s_1}$ for all $s_1, s_2 \geq 0$ on the interior $\widetilde{\mathcal{M}}_{flow}^{\circ}$). Consequently, on the manifold interior where bounding constraints remain slack, the terminal flow state depends solely on the net integrated control amounts $(s_1, s_2, s_3, s_4) = \int_0^T u(t)dt$ and is strictly invariant to the sequential order in which infinitesimal operations are executed. When active boundary hypersurfaces are encountered (such as the MTU envelope $g_1(x) = 1500x_2 - x_4 = 0$), control admissibility $\mathcal{U}(x)$ restricts admissible directions, requiring interleaving chaff packets ($u_3 > 0$) to expand total packet capacity before additional payload padding ($u_2 > 0$) can be absorbed.

Proof of Theorem 1. Let Ψ be the constraint residual defined in Section 3.1 of the main paper. Differentiating $\Psi_1(x(t))$ along the trajectory $\dot{x}(t) = \sum_{k=1}^{4} u_k(t)\vec{v}_k(x(t))$:

$$\begin{aligned}
\frac{d}{dt}\Psi_1(t) &= \dot{x}_8 - \frac{10^6(\dot{x}_4 + \dot{x}_5)}{x_6} + \frac{10^6(x_4 + x_5)\dot{x}_6}{x_6^2} \\
&= \left(-u_1\frac{x_8}{x_6} + u_2\frac{10^6}{x_6} + u_3\frac{L_0 \cdot 10^6}{x_6}\right) - \frac{10^6(u_2 + L_0 u_3)}{x_6} + \frac{10^6(x_4 + x_5)u_1}{x_6^2} \\
&= -\frac{u_1(t)}{x_6(t)}\left(x_8 - \frac{10^6(x_4 + x_5)}{x_6}\right) = -\frac{u_1(t)}{x_6(t)}\Psi_1(t)
\end{aligned} \tag{6}$$

Identical algebraic cancellation yields:

$$\frac{d}{dt}\Psi_2(t) = -\frac{u_1(t)}{x_6(t)}\Psi_2(t), \quad \frac{d}{dt}\Psi_3(t) = -\frac{u_3(t)}{x_2(t)}\Psi_3(t) \tag{7}$$

where the Ψ_3 identity follows by substituting $\dot{x}_{10} = (u_2 + u_3(L_0 - x_{10}))/x_2$, $\dot{x}_4 = u_2 + L_0 u_3$ and $\dot{x}_2 = u_3$ into $\dot{x}_{10} - \dot{x}_4/x_2 + x_4\dot{x}_2/x_2^2$. Hence, the constraint error vector $\Psi(t)$ satisfies the uncoupled homogeneous linear differential system $\dot{\Psi}(t) = A(t)\Psi(t)$, where $A(t) = \mathrm{diag}(-u_1(t)/x_6(t), -u_1(t)/x_6(t), -u_3(t)/x_2(t))$. Since $x_6(t) \geq x_6(0) > 0$ and $x_2(t) \geq x_2(0) \geq 1$, $A(t)$ is locally Lebesgue integrable ($A \in L^1_{\mathrm{loc}}$). Because $x_0 \in \widetilde{\mathcal{M}}_{\mathrm{flow}}$, the initial condition is $\Psi(0) = \mathbf{0}$. By the Carathéodory existence and uniqueness theorem for linear differential equations (Carathéodory, 1918), the unique absolutely continuous solution is $\Psi(t) \equiv \mathbf{0}$ for all $t \geq 0$.

Furthermore, examining the base physical coordinate trajectories across the boundary faces of $\widetilde{\mathcal{M}}_{\mathrm{flow}}$:

- $\dot{x}_0 = \dot{x}_1 = \dot{x}_3 = \dot{x}_5 \equiv 0$, satisfying strict immutability for $\mathcal{I}_{\mathrm{imm}}$.
- $\dot{x}_6 = u_1(t) \geq 0 \implies x_6(t) \geq x_6(0) > 0$, and $x_6(t) \leq 1.1 \times 10^8$ by the cap rule $u_1 = 0$ at the timeout boundary.
- $\dot{x}_2 = u_3(t) \geq 0 \implies x_2(t) \geq x_2(0) \geq 1$.
- **Total Bytes and Header Consistency ($x_4 \geq x_{11} + 20x_2$):** Chaff packets carry zero application payload; each chaff packet adds exactly 20 bytes of IPv4 header plus transport header H_0 ($H_0 = 20$ for minimal TCP, 32 with TSOPT, 8 for UDP), so its total packet size is $L_0 = 20 + H_0$. Differentiating the gap function $h(x) = x_4 - x_{11} - 20x_2$:

$$\begin{aligned} \dot{h}(x(t)) &= \dot{x}_4 - \dot{x}_{11} - 20\dot{x}_2 = (u_2 + L_0 u_3) - H_0 u_3 - 20u_3 \\ &= u_2(t) + (L_0 - H_0 - 20)u_3(t) = u_2(t) \geq 0 \end{aligned} \tag{8}$$

  Because $\dot{h}(t) = u_2(t) \geq 0$, the inequality $x_4(t) \geq x_{11}(t) + 20x_2(t)$ is monotonically non-decreasing along all Lie trajectories.

- **Transport Header Bounds ($H_{\min}x_2 \leq x_{11} \leq H_{\max}x_2$):** For TCP ($x_1 = 6$), $H_0 \in [20, 60]$, yielding $\dot{x}_{11} = H_0 u_3(t) \in [20, 60]\dot{x}_2(t)$, ensuring $20x_2(t) \leq x_{11}(t) \leq 60x_2(t)$ is forward-invariant; for UDP ($x_1 = 17$), $H_0 = 8$ strictly, maintaining $x_{11}(t) = 8x_2(t)$ identically.
- **Nagumo-Brézis Viability Condition for Manifold with Boundary and Corners:** The viable continuous domain $\widetilde{\mathcal{M}}_{\mathrm{flow}}$ is defined by the intersection of smooth

bounding hypersurfaces $g_i(x) \geq 0$:

$$g_1(x) = 1500x_2 - x_4 \geq 0 \quad \text{(MTU bound)} \tag{9}$$
$$g_2(x) = x_4 - x_{11} - 20x_2 \geq 0 \quad \text{(Header + payload lower bound)} \tag{10}$$
$$g_3(x) = x_{11} - x_2(20 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}) \geq 0 \tag{11}$$
$$g_4(x) = x_2(60 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}) - x_{11} \geq 0 \tag{12}$$
$$g_5(x) = x_6 - 1 \geq 0 \quad (x_{6,\min} = 1\,\mu\text{s}) \tag{13}$$
$$g_6(x) = x_2 - 1 \geq 0 \tag{14}$$
$$g_7(x) = 1.1 \times 10^8 - x_6 \geq 0 \quad \text{(flow-timeout cap)} \tag{15}$$

Because all constraint bounding functions $g_1(x), \ldots, g_7(x)$ are affine, the Bouligand contingency (tangent) cone $T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x)$ at any boundary or corner point coincides exactly with the intersection of the supporting half-spaces:

$$T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x) = \bigcap_{i:g_i(x)=0} \{v \in \mathbb{R}^{12} \mid \nabla g_i(x) \cdot v \geq 0\} \tag{16}$$

On any active boundary face where $g_i(x) = 0$, the inward normal velocity satisfies $\nabla g_i(x) \cdot \dot{x}(t) \geq 0$: $\nabla g_1 \cdot \dot{x} = (1500 - L_0)u_3(t) - u_2(t) \geq 0$ under admissible controls; $\nabla g_2 \cdot \dot{x} = u_2(t) \geq 0$; $\nabla g_3 \cdot \dot{x} = (H_0 - H_{\min})u_3(t) \geq 0$; $\nabla g_4 \cdot \dot{x} = (H_{\max} - H_0)u_3(t) \geq 0$; $\nabla g_5 \cdot \dot{x} = u_1(t) \geq 0$; $\nabla g_6 \cdot \dot{x} = u_3(t) \geq 0$; $\nabla g_7 \cdot \dot{x} = -u_1(t) \geq 0$ on $\{x_6 = 1.1 \times 10^8\}$, because the admissibility rule forces $u_1(t) = 0$ there. At boundary corners (intersections of multiple active faces), $\dot{x}(t) \in T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x)$. By the Nagumo-Brézis viability theorem (Nagumo, 1942; Brezis, 1970), $\widetilde{\mathcal{M}}_{\text{flow}}$ is forward-invariant under the dynamical system.

- **TCP Window Trajectory:** $\dot{x}_7 = u_4(t)\mathbb{I}_{x_1=6}\,\text{sat}((W_{\text{target}} - x_7)/\epsilon_{\text{win}})$. For UDP ($x_1 = 17$), $\vec{v}_{\text{win}} \equiv \mathbf{0}$, ensuring $x_7(t) \equiv 0$. For TCP ($x_1 = 6$), $x_7(t)$ moves monotonically toward $W_{\text{target}} \in [0, 65535]$ without crossing it, so it stays between $x_7(0)$ and W_{target} and hence within $[0, 65535]$ for all $t \geq 0$.

- **Global Existence and Uniqueness:** Because denominators $x_6(t) \geq x_6(0) > 0$ and $x_2(t) \geq x_2(0) \geq 1$ are bounded away from zero, the vector fields $\sum_{k=1}^{4} u_k(t)\vec{v}_k(x)$ are Carathéodory functions with linear growth. By Carathéodory's existence and uniqueness theorem (Carathéodory, 1918), solutions exist uniquely as absolutely continuous trajectories globally on $[0, \infty)$.

□

S2. Auxiliary algorithms

Algorithm S4 gives the general per-round Lie-guided adversarial training scheme for tree ensembles. The experiments in Section 5 of the main paper run a single-round instance of it.

Algorithm S1 Adaptive Directional Lie Derivative Estimator

Require: Classifier F, current flow state x, feature schema $\mathcal{S}$
Ensure: List of candidate evaluation tuples $\mathcal{D} = \{(k, \nabla_k, \vec{v}_k, \Delta s_k, f_k)\}_{k=1}^{4}$ (with $f_k = F(x_{\text{test}})$ the probed score), total exploratory queries q

$\mathcal{D} \leftarrow \emptyset$
$q \leftarrow 0$ {Counter for candidate exploratory probes}
{1. Evaluate Temporal Dilation Generator}
$\vec{v}_1 \leftarrow \vec{v}_{\text{time}}(x)$
$\Delta s_1 \leftarrow \max(1000.0, x_6 \cdot 0.50)$ {Adaptive temporal displacement in μs}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_1 \cdot \vec{v}_1, \mathcal{S})$
$\Delta t_1 \leftarrow \Delta s_1 / 10^6$ {Convert duration step to seconds}
$f_1 \leftarrow F(x_{\text{test}})$; $\nabla_1 \leftarrow (f_1 - F(x))/\Delta t_1$ {Derivative per second of delay}
Append $(1, \nabla_1, \vec{v}_1, \Delta s_1, f_1)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{2. Evaluate Payload Padding Generator}
$\vec{v}_2 \leftarrow \vec{v}_{\text{pad}}(x)$
$\Delta s_2 \leftarrow \max(50.0, x_4 \cdot 0.25)$ {Adaptive forward byte padding step}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_2 \cdot \vec{v}_2, \mathcal{S})$
$f_2 \leftarrow F(x_{\text{test}})$; $\nabla_2 \leftarrow (f_2 - F(x))/\Delta s_2$
Append $(2, \nabla_2, \vec{v}_2, \Delta s_2, f_2)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{3. Evaluate Chaff Packet Injection Generator}
$\vec{v}_3 \leftarrow \vec{v}_{\text{chaff}}(x)$
$\Delta s_3 \leftarrow \max(1.0, \text{round}(x_2 \cdot 0.50))$ {Adaptive chaff packet injection count}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_3 \cdot \vec{v}_3, \mathcal{S})$
$f_3 \leftarrow F(x_{\text{test}})$; $\nabla_3 \leftarrow (f_3 - F(x))/\Delta s_3$
Append $(3, \nabla_3, \vec{v}_3, \Delta s_3, f_3)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{4. Evaluate TCP Window Morphing Generator}
if $x[1] = 6$ **then**
    {TCP flows only; window size is inapplicable to UDP}
    $\vec{v}_4 \leftarrow \vec{v}_{\text{win}}(x)$
    $\Delta s_4 \leftarrow |64240.0 - x[7]|$
    if $\Delta s_4 > 0$ **then**
        $x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_4 \cdot \vec{v}_4, \mathcal{S})$
        $f_4 \leftarrow F(x_{\text{test}})$; $\nabla_4 \leftarrow (f_4 - F(x))/(\Delta s_4/65535.0)$ {Normalized window gradient}
        Append $(4, \nabla_4, \vec{v}_4, \Delta s_4, f_4)$ to $\mathcal{D}$
        $q \leftarrow q + 1$
    end if
end if
return $\mathcal{D}, q$

S3. Per-family results

Table S1 breaks the XGBoost results of the main text down by attack family (all families that appear in the 1,000-flow cohort; families with a handful of flows, such as one CIC-IDS2017 botnet flow, give no reliable rate). On CSE-CIC-IDS2018 the families in the cohort are evaded in 98.7%–100.0% of the flows (DoS 390 flows, DDoS 375, brute force 200, botnet 35). On CIC-IDS2017 DoS (422 flows) is evaded in 98.1% and the other families in 100.0%. On UNSW-NB15 more than half of the cohort are exploits, which are evaded in only 44.6% of 504 flows and so set the aggregate figure; generic attacks (50.0% on 22 flows) and DoS

Algorithm S2 Analytical Manifold Projection Operator $\mathcal{P}_{\mathcal{M}}$

Require: Perturbed vector $x \in \mathbb{R}^{12}$, feature schema $\mathcal{S}$, optional baseline flow x_{orig}
Ensure: Projected flow vector $x_{\text{proj}} \in \mathcal{M}_{\text{flow}}$
$x_{\text{proj}} \leftarrow \text{copy}(x)$
if x_{orig} is provided **then**
  $x_{\text{proj}}[j] \leftarrow x_{\text{orig}}[j]$ for all $j \in \mathcal{I}_{\text{immut}}$ {Preserve immutable features}
  $x_{\text{proj}}[j] \leftarrow \max(x_{\text{orig}}[j], x_{\text{proj}}[j])$ for all $j \in \mathcal{I}^{+}_{\text{mono}}$ {Enforce monotonicity lower bounds}
end if
{Enforce discrete bounds and protocol-dependent packet/header sizes}
$x_{\text{proj}}[2] \leftarrow \max(1.0, \text{round}(x_{\text{proj}}[2]))$ {Integer Forward Packets}
$x_{\text{proj}}[3] \leftarrow x[3]$ {Immutable Backward Packets Preserved}
$H_{\min} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 20.0 **else** 8.0 {Min transport header: 20B TCP, 8B UDP}
$H_{\max} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 60.0 **else** 8.0 {Max transport header: 60B TCP, 8B UDP}
$x_{\text{proj}}[11] \leftarrow \min(x_{\text{proj}}[2] \times H_{\max}, \max(x_{\text{proj}}[2] \times H_{\min}, \text{round}(x_{\text{proj}}[11])))$ {Transport Header Length}
$x_{\text{proj}}[4] \leftarrow \min(x_{\text{proj}}[2] \times 1500.0, \max(0.0, \text{round}(x_{\text{proj}}[4])))$ {Forward Bytes $\in [0, 1500N_{\text{fwd}}]$}
$x_{\text{proj}}[5] \leftarrow x[5]$ {Immutable Backward Bytes Preserved}
$x_{\text{proj}}[6] \leftarrow \min(1.1 \times 10^8, \max(1.0, \text{round}(x_{\text{proj}}[6])))$ {Integer Duration $\geq 1\mu$s, capped below flow timeout}
$x_{\text{proj}}[7] \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** $\text{clamp}(\text{round}(x_{\text{proj}}[7]), 0.0, 65535.0)$ **else** 0.0 {TCP Window $[0, 65535]$, 0 for UDP}
{Analytically recalculate coupled non-linear flow rates}
$T_{\text{sec}} \leftarrow x_{\text{proj}}[6]/10^6$
$B_{\text{tot}} \leftarrow x_{\text{proj}}[4] + x_{\text{proj}}[5]$
$N_{\text{tot}} \leftarrow x_{\text{proj}}[2] + x_{\text{proj}}[3]$
$x_{\text{proj}}[8] \leftarrow B_{\text{tot}}/T_{\text{sec}}$ {Flow Bytes/s}
$x_{\text{proj}}[9] \leftarrow N_{\text{tot}}/T_{\text{sec}}$ {Flow Packets/s}
$x_{\text{proj}}[10] \leftarrow x_{\text{proj}}[4]/x_{\text{proj}}[2]$ {Forward Packet Length Mean}
return x_{proj}

(51.2% on 80) are similar, whereas fuzzers (79.5% on 219) and reconnaissance (69.2% on 143) are easier.

S4. Ablations

Table S2 compares FFE with and without tree-split boundary snapping (TSBS) on the full 1,000-flow cohorts of the three corpora. Table S3 reports the other ablations on the CUDA XGBoost target of CSE-CIC-IDS2018 with a balanced cohort of 283 detected malicious flows (up to 50 per attack family): Panel A varies the step budget M, Panel B isolates generator subsets, and Panel C compares operational cost regimes. The TSBS relative snapping offset ϵ_{rel} was swept over $\{0, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$.

Step budget. ASR rises from 42.8% at $M = 1$ (5.0 queries) through 67.1% at $M = 3$ and 84.1% at $M = 5$ to 98.9% at $M = 10$ and $M = 20$ (14.4 queries at $M = 20$). Physical feasibility is 100.0% for every budget.

Generators. No single generator suffices. Temporal dilation alone evades 66.1% of the flows but needs 32.6 queries and adds 40.7 s of flow duration on average; window morphing alone evades 46.3% with 2.5 queries and no added delay; chaff alone 32.5%; padding alone 10.2%. Pairs are better than singles (time plus chaff 78.1%, time plus padding 69.3%, padding plus chaff 33.6% with no added delay), and the complete cone reaches 98.9% with 14.4 queries and 0.18 s of mean delay. The full cone needs fewer queries than the time-based

Table 1: *
Table S1. FFE against XGBoost by attack family.

Dataset	**Attack family**	n	**Real ASR (%)**	**Queries**
CSE-CIC-IDS2018	DoS	390	100.0	7.6
	DDoS	375	98.7	29.2
	BruteForce	200	100.0	9.9
	Botnet	35	100.0	18.4
CIC-IDS2017	DoS	422	98.1	8.0
	PortScan	322	100.0	29.4
	DDoS	235	100.0	5.0
	BruteForce	20	100.0	28.0
	Botnet	1	100.0	9.0
UNSW-NB15	Exploits	504	44.6	116.7
	Fuzzers	219	79.5	56.4
	Reconnaissance	143	69.2	69.4
	DoS	80	51.2	104.7
	Generic	22	50.0	117.2
	Shellcode	19	100.0	13.8
	Backdoor	6	66.7	103.5
	Worms	6	33.3	131.0
	Analysis	1	100.0	9.0

Table 2: *
Table S2. FFE with and without tree-split boundary snapping (XGBoost, Real ASR with Wilson 95% interval).

Dataset	**Variant**	**Real ASR (%)**	**Queries**	**ms/flow**
CSE-CIC-IDS2018	FFE	99.5 [98.8, 99.8]	16.5	15.8
	FFE without TSBS	99.5 [98.8, 99.8]	11.6	10.6
CIC-IDS2017	FFE	99.2 [98.4, 99.6]	14.6	10.7
	FFE without TSBS	97.5 [96.3, 98.3]	13.5	9.6
UNSW-NB15	FFE	57.6 [54.5, 60.6]	93.7	89.1
	FFE without TSBS	54.6 [51.5, 57.7]	51.5	36.0

Algorithm S3 TSBS Candidate Generation (GenerateCandidates)

Require: Current flow $x^* \in \mathcal{M}_{\text{flow}}$, active splits active_splits (pairs (j, θ_j)), tolerance ϵ_{rel}
Ensure: Candidate boundary-crossing pool $\mathcal{C}$ (before manifold projection)

$\mathcal{C} \leftarrow \emptyset$
{1. Monotonic features: closest split strictly above $x^*[j]$; UDP header x_{11} is immutable at 8B}
for each $j \in \mathcal{I}^+_{\text{mono}}$ **with** NOT $(j = 11 \wedge x^*[1] = 17)$ **do**
  $\theta^*_j \leftarrow \min\{\theta_j \mid (j, \theta_j) \in \text{active_splits} \ \wedge \ \theta_j > x^*[j]\}$
  if θ^*_j exists **then**
    $\epsilon_j \leftarrow 1.0$ **if** $j \in \{2, 4, 11\}$ **else** $\text{round}(\epsilon_{\text{rel}} \cdot \max(1.0, |\theta^*_j|))$
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[j] \leftarrow \theta^*_j + \epsilon_j$
    if $j = 11$ **then**
      $x_c[11] \leftarrow \min(60.0 \cdot x_c[2], x_c[11])$
      $x_c[4] \leftarrow \min(1500.0 \cdot x_c[2], \max(0.0, x_c[4]))$
    end if
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{2. Invert coupled rate splits (x_8, x_9) into duration dilation}
for each $(r, a, b) \in \{(8, x^*[4], x^*[5]), (9, x^*[2], x^*[3])\}$ **do**
  $\theta^*_r \leftarrow \max\{\theta_r \mid (r, \theta_r) \in \text{active_splits} \ \wedge \ 0 < \theta_r < x^*[r]\}$
  if θ^*_r exists **and** $d \leftarrow 10^6(a + b)/\theta^*_r + 10.0$ satisfies $d \leq 1.1 \times 10^8$ **then**
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[6] \leftarrow d$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{3. Invert coupled packet length splits (x_{10}) into chaff packet injection}
$H_0 \leftarrow 32.0$ **if** $(x^*[1] = 6 \wedge x^*[11]/x^*[2] \geq 26.0)$ **else** 20.0 **if** $x^*[1] = 6$ **else** 8.0
$L_0 \leftarrow 20.0 + H_0$ {52B/40B TCP (with/without RFC 7323 options), 28B UDP}
$\theta^*_{10} \leftarrow \max\{\theta_{10} \mid (10, \theta_{10}) \in \text{active_splits} \ \wedge \ L_0 < \theta_{10} < x^*[10]\}$
if θ^*_{10} exists **then**
  $N \leftarrow \lfloor (x^*[4] - \theta^*_{10} x^*[2])/(\theta^*_{10} - L_0) \rfloor + 1$
  if $N \leq 1000$ **then**
    {Chaff budget cap: $N \leq 1000$ packets}
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[2] \leftarrow x^*[2] + N$
    $x_c[4] \leftarrow x^*[4] + N \cdot L_0$; $x_c[11] \leftarrow x^*[11] + N \cdot H_0$
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end if
{4. TCP window morphing to benign OS baseline}
if $x^*[1] = 6 \ \wedge \ x^*[7] \neq 64240.0$ **then**
  $x_c \leftarrow \text{copy}(x^*)$; $x_c[7] \leftarrow 64240.0$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
end if
return $\mathcal{C}$

subsets, possibly because the cheap window and chaff moves are available when delay alone would stall.

Cost regimes and snapping offset. These do not discriminate on this corpus. All four cost regimes reach 98.9% ASR with 14.1–14.4 queries, and the snapping offset leaves ASR at 98.9% and the queries between 14.2 and 14.4. The regimes do change the mean added delay (181 ms under regime A, 133 ms under the latency-sensitive regime B), so FFE responds to the operator's cost model, but because the corpus is saturated we have no evidence on how the regime affects success; a harder target such as UNSW-NB15 would be needed for that.

Table 3: *

Table S3. Ablation of FFE on CUDA XGBoost, CSE-CIC-IDS2018 ($N = 283$). Delay is the mean added flow duration.

Configuration	**ASR (%)**	**PFR (%)**	**Queries**	**Added delay (s)**
Panel A: step budget M				
$M = 1$	42.8	100.0	5.0	0.08
$M = 2$	63.3	100.0	8.3	0.14
$M = 3$	67.1	100.0	9.5	0.15
$M = 5$	84.1	100.0	12.2	0.16
$M = 10$	98.9	100.0	13.9	0.18
$M = 20$	98.9	100.0	14.4	0.18
$M = 50$	98.9	100.0	16.0	0.18
Panel B: generator subset ($M = 20$)				
Temporal Dilation Only	66.1	100.0	32.6	40.72
Payload Padding Only	10.2	100.0	5.1	0.00
Chaff Injection Only	32.5	100.0	9.7	0.00
Window Morphing Only	46.3	100.0	2.5	0.00
Time + Padding	69.3	100.0	39.6	26.81
Time + Chaff	78.1	100.0	38.4	22.05
Padding + Chaff	33.6	100.0	14.2	0.00
Complete Lie Cone	98.9	100.0	14.4	0.18
Panel C: operational cost regime ($M = 20$)				
A (Default / Balanced)	98.9	100.0	14.4	0.18
B (Latency-Sensitive)	98.9	100.0	14.1	0.13
C (Unconstrained Delay)	98.9	100.0	14.4	0.18
D (Uniform / Flat)	98.9	100.0	14.3	0.14

Algorithm S4 Lie-Guided Adversarial Training (LGAT) for Tree Ensembles

Require: Training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, number of boosting rounds T, adversarial ratio $\beta = 0.5$, step budget $M_{\text{train}} = 5$
Ensure: Robust tree ensemble $F_T(x)$

Initialize base tree $F_0 \leftarrow \text{FitCleanBaseTree}(\mathcal{D}_{\text{train}})$
for $t = 1$ **to** T **do**
  $\mathcal{D}_{\text{aug}} \leftarrow \emptyset$
  for each $(x_i, y_i) \in \mathcal{D}_{\text{train}}$ **do**
    if $y_i = 1$ **and** $\text{Uniform}(0, 1) < \beta$ **then**
      $x_{\text{adv}} \leftarrow \text{FFE}(F_{t-1}, x_i, \text{budget} = M_{\text{train}})$
      Add $(x_{\text{adv}}, 1)$ to $\mathcal{D}_{\text{aug}}$
    else
      Add (x_i, y_i) to $\mathcal{D}_{\text{aug}}$
    end if
  end for
  Compute negative gradients g_i and Hessians h_i on $\mathcal{D}_{\text{aug}}$
  Fit tree f_t using robust split scoring $\text{Gain}_{\text{robust}}$
  $F_t(x) \leftarrow F_{t-1}(x) + \eta \cdot f_t(x)$
end for
return F_T

S5. Cross-architecture transferability

Flows crafted by FFE against a source ensemble were scored by the three other architectures (Table S4). The cohort is the 221 CSE-CIC-IDS2018 malicious flows (up to 50 per family) that XGBoost, LightGBM, CatBoost and Random Forest all detect when clean, so a transferred evasion is not the target's own false negative; all four models are trained on the same training partition. Flows crafted against XGBoost evade LightGBM in 71.5% of the cases, Random Forest in 55.7% and CatBoost in 24.0%. LightGBM flows evade XGBoost in 75.6%, Random Forest in 63.8% and CatBoost in 24.0%; CatBoost flows evade XGBoost in 70.1%, LightGBM in 77.4% and Random Forest in 77.4%; Random Forest flows evade XGBoost in 78.7%, LightGBM in 71.0% and CatBoost in 24.0%. Excluding CatBoost as target, 55.7%–78.7% of the transferred flows evade. CatBoost is the hardest target for the three other sources, whose flows each evade it in 53 of 221 cases (we did not check whether these are the same flows); attacks against CatBoost itself succeed in only 69.7% of the cases. We did not test why CatBoost differs; its symmetric oblivious trees place thresholds differently from the leaf-wise and depth-wise ensembles.

Table 4: *

Table S4. Transferability of FFE flows (attack success rate in %, $N = 221$; diagonal: attack against the source model itself).

Source \ Target	XGBoost (CUDA)	LightGBM	CatBoost	Random Forest
XGBoost (CUDA)	**100.0**	71.5	24.0	55.7
LightGBM	75.6	**100.0**	24.0	63.8
CatBoost	70.1	77.4	**69.7**	77.4
Random Forest	78.7	71.0	24.0	**100.0**

S6. Corpus statistics

Table S5 gives the record counts of the three corpora. Valid records pass the validity filter; Kept = Valid − Duplicates, where duplicates are records with identical features and label after projection onto the flow manifold. Records with identical features but different labels are kept (149,526, 22 and 1,040).

Table 5: *
Table S5. Full-scale benchmark corpora.

Dataset	Raw rows	Valid	Duplicates	Kept	Attacks	Train	Test
CSE-CIC-IDS2018	8,284,254	8,153,708	3,185,319	4,968,389	594,809	3,477,872	993,679
CIC-IDS2017	2,099,971	2,085,851	426,982	1,658,869	396,403	1,161,208	331,774
UNSW-NB15	257,673	215,689	92,181	123,508	55,559	86,456	24,703

S7. White-box leaf-tuple-style control

The official LT-Attack code (chong-z/tree-ensemble-attack) needs Bazel and a C++ compiler, which this machine lacks, so we wrote a greedy attack in the same spirit (`src/attacks/whitebox_tree`). It reads all 60 XGBoost trees; each step moves one feature to the other side of one split threshold, choosing the move that lowers the exact ensemble margin most, for at most 20 steps (margins agree with `predict_proba` to 2×10^{-7}). It is greedy and not LT-Attack's leaf-tuple search with an L_∞ objective. Two variants end in the same projection and bookkeeping as the other attacks: *project at end* (unconstrained attack, then projection, then re-scoring) and *project in loop* (every candidate is projected before scoring). Both use all 1,000 cohort flows of the main benchmark, two target queries (the tree reads are not counted; about 300–1,000 internal margin evaluations per flow) and are schema-feasible by construction.

Table 6: *
Table S7. Real ASR (%) of the white-box control and FFE against XGBoost; exact McNemar p against FFE on the same 1,000 flows.

Corpus	Project at end	Project in loop	FFE	p (in loop vs FFE)
CSE-CIC-IDS2018	31.6	99.5	99.5	1.0
CIC-IDS2017	35.1	100.0	99.2	0.008
UNSW-NB15	0.0	90.2	57.6	10^{-87}

Before projection the attack evades every flow on all three corpora; most of that comes from the derived packet-rate columns and the immutable columns, which the projection undoes. The projected-at-end variant is the fair version of an unconstrained white-box attack and is far below FFE ($p < 10^{-170}$). The in-loop variant equals FFE on CSE-CIC-IDS2018, beats it on the other two corpora (on CIC-IDS2017 by eight flows against none), and on UNSW-NB15 by 32 points. It sees the trees and is cost-blind: its median operational cost is 213 against 5.6 for FFE on UNSW-NB15. FFE's four generators therefore limit its reach on

UNSW-NB15, and its advantage is that it needs only scores (FFE without snapping) and its cost, not the success rate; the control counts two queries and reads the trees for free, whereas FFE counts every probe.

S8. Packet-level replay

We built a synthetic replay (`src/validation/packet_replay.py`); no packet capture was used and nothing was sent over a network. A flow record is expanded into a packet sequence with the stated packet counts, payload and header totals, duration and initial window, and a CICFlowMeter-style extractor (our re-implementation, not CICFlowMeter) recomputes the twelve features. For 20,000 flows per corpus the round trip is exact for every flow on CSE-CIC-IDS2018 and CIC-IDS2017 and for 19,999 of 20,000 on UNSW-NB15 (one one-packet record with a duration of 1 μs). This shows that the schema is jointly realisable by packet lists, not that it matches a real capture.

We then derived the delay, chaff count, padding and window from each FFE adversarial flow (first 300 cohort flows per corpus, XGBoost; schema-level success 99.3%, 99.3% and 59.7%), applied the four actions to the packet trace, re-extracted the features and re-scored. The coupled rates, forward mean, duration and window match the schema vector to 10^{-6}; only the forward header length disagrees, on 9.1%, 2.3% and 6.1% of the successful flows (up to 62% on single flows), because snapping adds packets and the projection fills headers at 20 B per packet. All re-extracted vectors pass the feasibility check. The generators book 28–52 B of forward payload per chaff packet, whereas a minimal packet carries none, so we replayed three conventions. Table S8 gives the share of evasions that remain evasive.

Table 7: *

Table S8. Share of schema-level evasions that remain evasive after packet-level replay (XGBoost).

Corpus	**Evasions**	**Chaff carries booked bytes**	**Empty chaff, bytes as padding**	**Er**
CSE-CIC-IDS2018	298	99.3	99.3	
CIC-IDS2017	298	100.0	100.0	
UNSW-NB15	179	100.0	100.0	

UNSW-NB15 depends on the booked chaff bytes because 69% of its successful flows use chaff and 86% use padding. The replay does not model a TCP state machine, server reaction, offload or flow timeouts, and part of its agreement holds by construction because the action magnitudes are derived from the adversarial vector. It does not test whether an attack payload still works after dilation or padding.

S9. Why LightGBM resists FFE on CIC-IDS2017

On the 1,000-flow cohort FFE evades XGBoost in 99.2% and LightGBM in 35.1%. By family, LightGBM is evaded in 100% of port scans, botnet and brute-force flows but in 1.9% of DoS

and 0.0% of DDoS flows, which are about 66% of the cohort; XGBoost is evaded in 98.1% and 100% of them. The threshold extraction and the probe are correct for LightGBM, start scores are identical (median 0.998) and the gain profiles are nearly the same (backward payload 0.31 for both, duration 0.29 for both), so reliance on immutable features does not explain the gap. Setting only the initial window to 64,240 B, FFE's window target, flips 97.2% of the 33,857 detected DoS test flows for XGBoost and 0.9% for LightGBM (99.96% against 0.0% for DDoS). No training flow has a window between 29,200 and 65,535 B; XGBoost splits at 32,831 and 65,535, so 64,240 B lands on the benign side, whereas LightGBM splits at 63,586 and 65,435 and keeps it on the malicious side. More leaves, finer histogram bins and a smaller minimum leaf size change this by about one point. Moving only the window target to the in-support value 65,535 B lifts LightGBM from 35.1% to 93.7% (queries 120 to 21.8) and leaves XGBoost at 99.2%. The gap is therefore a property of one generator's fixed target meeting an empty region of the data, not evidence that LightGBM is more robust; we did not examine the remaining DoS failures at 65,535 B.

S10. Related-work taxonomy

Table 8: *

Table S10. Representative evasion, verification and defense paradigms for tabular and network telemetry. Physical feasibility reports the empirical PFR of this benchmark where it was evaluated (main-text results section).

Method	**Target model**	**Engine**	**Queries**	**Coupled invariants**	**Physical feasibility**
Kantchelian et al.	GBDT, random forest	MILP (worst case $2^{\mathcal{O}(DT)}$)	0 (white-box)	Unsupported (linear only)	Unconstrained
Zhang et al. (LT-Attack)	Tree ensembles	Discrete leaf-tuple search	$O(N_{\text{iter}})$	Not modeled	Unconstrained
Chen et al. (ZOO)	Black-box classifiers	Zeroth-order coordinate descent	80–130 (mean)	Not modeled	0.0–83.2% of evasive flows (empirical)
Chen et al. (HSJ)	Black-box decision	Boundary binary walk	47 (mean)	Not modeled	0.0% (empirical)
Lin et al. (IDSGAN); Zolbayar et al.	Black-box NIDS	Learned generator on a surrogate	2.4–4.5 after training	Repaired after generation	100.0% (after repair, our re-implementation)
Ghamizi et al. (MOEVA)	Constrained tabular	Multi-objective genetic search	226–445 (mean)	Heuristic penalty	59.4–100.0% of evasive flows (empirical)
Simonetto et al. (CAA/CAPGD)	Tabular models	Constrained APGD (surrogate) with MOEVA	3.2–16.2 (CAPGD, mean)	Repair operator	100.0% (after repair, our re-implementation)
PSO baseline after Han et al. (ours)	Black-box NIDS	PSO on interval and padding features	183–251 (mean)	Invariant projection	100.0% (projected, our addition)
Flow-Field Evasion (ours)	**XGBoost, LightGBM**	**Lie vector field steering**	$\leq 4M_{\max}+1$ **probes + TSBS (mean 14.6–93.7)**	**Exact closed-form**	**100.0% (schema-feasible)**

Note: T trees, D depth, $M_{\max} = 20$, $K = 4$ generators. Empirical query counts and feasible shares of the evasive candidates are ranges over the three full-scale benchmarks against XGBoost; FFE and the white-box rows count their own probes (main-text Table 1). LT-Attack was not run; Section S7 reports a greedy leaf-tuple-style control. CAPGD and the NIDSGAN-style generator are our re-implementations.